Writing Assignment #11: Functions, Inverses, and Modular Arithmetic

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Section 4.2

Question 1

Part D

The sine function is not bijective as it is not one to one. However, over a closed domain from $\frac{-\pi}{2}, \frac{\pi}{2}$ it does work because there are no repeated outputs. The inverse would be the arcsin of x when restricted.

Part F

This function can be simplified to 1-$\frac{1}{x}$ in order to find the inverse.

$y - 1 = -\frac{1}{x} \rightarrow 1 - y = \frac{1}{x} \rightarrow x = \frac{1}{1-y}$

This is a function, for all x not equal to 0, as it is bijective.

Question 12

Claim: $\exists f \ni Dom(D) \wedge D \subseteq A \wedge f = g|_D \wedge \forall x, i(x) = x \Rightarrow f = g \circ i$

Then, $(\forall x \in D) \ni f(x) = g(x)$

Thus, $(\forall x \in i(x) = x) \ni g(i(x)) = g(x)$

So, f(x)=g(i(x))

Hence, f(x)=(g $\circ$i)(x)

Then, $f = g \circ i|_D$,

Thus, f=g $\circ$i

$\therefore \exists f \ni Dom(D) \wedge D \subseteq A \wedge f = g|_D \wedge \forall x, i(x) = x \Rightarrow f = g \circ i$

this feels really messy but I didn't really know what else to do here, I think this works?

Question 15

Part A

Claim: $f : A \to B \wedge g : C \to D \wedge f \times g = \{(a, c), (b, d) : (a, b) \in f \wedge (c, d) \in g\} \Rightarrow f \times g$ *is a mapping from* $A \times C$ *to* $B \times D$

Suppose $(a, c) \in A \times C$:

Then, $(\exists b \in B)(\exists d \in D) \ni (a, b) \in f \wedge (c, d) \in g$

Thus, ((a,c),(b,d)) $\in f \times g$

So, $(\forall (a, c) \in A \times C)(\exists (b, d) \in B \times D) \ni ((a, c), (b, d)) \in f \times g$

$\therefore f : A \to B \wedge g : C \to D \wedge f \times g = \{(a, c), (b, d) : (a, b) \in f \wedge (c, d) \in g\} \Rightarrow f \times g$ *is a mapping from* $A \times C$ *to* $B \times D$

Part B

(f(a),g(c))

Question 19

Part B

This proof gets a grade of A.

Part C

This proof gets a grade of F because it assumes that g $\circ$f = f $\circ$g which is not always true.

Section 4.3

Question 3

Part B

This function is surjective as each y cooresponds to an x but it is not injective because multiple integers can map to $\mathbb{Z}_5$

Part C

Same type of logic applies to this one. This is surjective as each output has an input but each output has duplicate inputs.

Question 9

Part B

$A = \{1\}, B = \{2, 3\}, C = \{4, 5\}$
f(1)=2, g(2)=4, g(3)=5
g ∘f(1)=g(2)=4

Part D

$A = \{1, 2\}, B = \{3, 4, 5\}, C = \{6\}$
f(1)=3, f(2)=4, g(3)=6, g(4)=6, g(5)=6
g ∘f(1)=g(3)=6, g ∘f(2)=g(4)=6

Question 12

Part B

Claim: f: $\mathbb{Z}_4 \to \mathbb{Z}_2$ given by $f(\overline{x}) = [3x]$ is onto but not one to one.
Evaluating:
$f(\overline{0}) = [0], f(\overline{1}) = [3] = [1], f(\overline{2}) = [6] = [0], f(\overline{3}) = [9] = [1]$
Onto:
Then, both $[0] \wedge [1] \in \mathbb{Z}_2$ are achieved
Thus, every element of the codomain has a preimage
So, f is onto
Not one to one:
Then, $f(\overline{0}) = f(\overline{2} = [0]$
So, f maps distinct elements to the same value
Hence, f is not one to one.
$\therefore f : \mathbb{Z}_4 \to \mathbb{Z}_2$ given by $f(\overline{x}) = [3x]$ is onto but not one to one.

Part D

Claim: f: $\mathbb{Z}_4 \to \mathbb{Z}_4$ given by $f(\overline{x}) = [2x]$ is neither onto nor not one to one.
Evaluating: $f(\overline{0}) = \overline{0}, f(\overline{1}) = \overline{2}, f(\overline{2}) = \overline{0} f(\overline{3}) = \overline{2}$
Not one to one:
Then, $f(\overline{0}) = f(\overline{2}) = \overline{0}$
Thus, distinct inputs have the same output
So, f isn't one to one
Not onto:
Then, the image of f is $\{\overline{0}, \overline{2}\}$
So, $\overline{1} \vee \overline{3}$ are not in the image
Thus, f isn't onto
$\therefore f : \mathbb{Z}_4 \to \mathbb{Z}_4$ given by $f(\overline{x}) = [2x]$ is neither onto nor not one to one.

Question 15

Part C

This proof gets a grade of C as it states a correct idea and process but fails to prove surjectivity as it doesn't refer to arbitrary elements in C.

Part G

This proves gets a grade of C as it incorrectly uses injectivity.